STAT-437: APPLIED STATISTICAL METHODS
Assignment: 3
Due: October 16, 2026 at 10:00PM
Total=40 Marks

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Instructions:(i) All answers and relevant R output must be provided in the main body of the assignment. Include all R code in the Appendix.
(ii) Submit the source code as an .R or .Rmd file.
(iii) A PDF version of the assignment is also required.
(iv) Clearly explain your statistical conclusions; simply reporting R output is not sufficient.
(v) Late submissions will not be accepted.

1 Ohio Children Wheeze Status

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"ohio" data set is available in "faraway" r-package. It is a subset of the Six-City Study, a longitudinal study investigating the health effects of air pollution.

- Description: The ohio data frame has 2148 observations and 4 variables.
- Usage: library(faraway) and data(ohio)
- Variables:
  resp: Indicator of wheeze status (1 = yes, 0 = no).
  id: Subject identification number.
  age: Age of the child, where age = 0 corresponds to 9 years old.
  smoke: Indicator of maternal smoking at the first year of the study.
- References:
  (i)Fitzmaurice, G.M. and Laird, N.M. (1993) A likelihood-based method for analyzing longitudinal binary responses, Biometrika 80: 141-151.

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Give the answers of the following:

1. Consider the following logistic Generalized Linear Mixed Effects Model:

$$log(\frac{\mu_{ij}}{1-\mu_{ij}}) = (\beta_0 + r_i) + \beta_1 age_{ij} + \beta_2 smoke_{ij} \tag{1}$$

where $\mu_{ij} = E(Y_{ij}|r_i) = P(Y_{ij} = 1|r_i)$ and $r_i \sim N(0, \sigma_r^2)$.

Fit this model using an appropriate R function for GLMMs, with id as the grouping variable.

(a) Write down the fitted model, including the estimated fixed-effect coefficients and the estimated standard deviation of the random intercept.

(b) Interpret the estimated coefficients for age and smoke in terms of the odds of wheezing, keeping the other covariate fixed.

(c) Explain what the standard deviation of the random intercept represents in this model. In particular, explain what it tells us about differences in the baseline probability of wheezing among children.

2. Now consider a model that allows the effect of maternal smoking to vary across children:

$$log(\frac{\mu_{ij}}{1-\mu_{ij}}) = (\beta_0 + r_{0i}) + \beta_1 age_{ij} + (\beta_2 + r_{1i}) smoke_{ij} \tag{2}$$

where $\mu_{ij} = E(Y_{ij}|r_{0i}, r_{1i}) = P(Y_{ij} = 1|r_{0i}, r_{1i})$ and $\binom{r_{0i}}{r_{1i}} \sim N[\binom{0}{0}, \sigma_r^2]$ with covariance matrix $\sigma_r^2 = \begin{pmatrix} \sigma_{r0}^2 & \sigma_{r01} \\ \sigma_{r01} & \sigma_{r1}^2 \end{pmatrix}$, $\sigma_{r0}^2 = Var(r_{0i}), \sigma_{r1}^2 = Var(r_{1i}), \sigma = Cov(r_{0i}, r_{1i})$.

Fit the model using a random intercept and random slope for smoke, with id as the grouping variable.

(a) Report the estimated fixed effects and the estimated random-effects standard deviations.

(b) Interpret the fixed-effect coefficients for age and smoke. Explain how the interpretation of the smoke coefficient differs from Question 1(b).

(c) Interpret the estimated standard deviation of the random effect associated with smoke. What does this tell us about the variability of the effect of maternal smoking across children?

(d) Report the estimated correlation between the random intercept and the random smoke effect. Explain what a positive or negative correlation would mean in this context.

2 Theoretical Questions: Linear and Generalized Linear Mixed Effects Models

3. (a) Compare the role of the link function in a Linear Mixed Model (LMM) and a Generalized Linear Mixed Model (GLMM). What is the role of the link function in a GLMM, and why is it generally not explicitly required in an LMM?

   (b) Why are random effects introduced in a Linear Mixed Effects (LME) model for longitudinal data? Explain the difference between fixed effects and random effects, and describe the interpretation of a random intercept for an individual.

   (c) What estimation methods are commonly used for estimating the parameters of Linear Mixed Effects Models (LMMs) and Generalized Linear Mixed Models (GLMMs)? Explain why estimation of GLMMs is more difficult than estimation of LMMs.

   (d) A researcher has binary longitudinal data with repeated observations for each individual. Explain why a standard logistic regression model may not be appropriate for these data. How does a GLMM address the dependence among repeated binary responses, and what assumption is made about the responses after conditioning on the random effects?